AQ8.3: Activity Questions 3 - Not Graded

This assignment will not be graded and is only for practice.

Level 1

1 point

Consider two vectors $a = (1, -2, 2), b = (4, 0, -3)$a = (1, −2, 2), b = (4, 0, −3). Using the standard dot product as the inner product, the projection of aa in the direction of bb is given by

○

$(\frac{-8}{25}, 0, \frac{6}{25})(25-8, 0, 256)$

○

$(\frac{-8}{25}, \frac{6}{25})(25-8, 256)$

○

$(\frac{8}{25}, 0, \frac{-6}{25})(258, 0, 25-6)$

○

$(\frac{8}{25}, \frac{-6}{25})(258, 25-6)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(\frac{-8}{25}, 0, \frac{6}{25})(25-8, 0, 256)$

1 point

Consider an orthonormal basis $\{\frac{1}{\sqrt{5}}(1, 2), \frac{1}{\sqrt{5}}(-2, 1)\}\{ \quad 5$

√

$1(1, 2), \quad 5$

√

$1(-2, 1)\}$ of R^2 R2. If $(3, 5) = c_1 v_1 + c_2 v_2$ (3, 5) = c1v1 + c2v2 where $c_1, c_2 \in R$c1, c2 ∈ R and $v_1 = \frac{1}{\sqrt{5}}(1, 2), v_2 = \frac{1}{\sqrt{5}}(-2, 1)$v1 = $\quad 5$

√

$1(1, 2)$, v2 = $\quad 5$

√

$1(-2, 1)$, then choose the correct option.

○

$c_1 = \frac{13}{\sqrt{5}}, c_2 = \frac{1}{\sqrt{5}}$c1 = $\quad 5$

√

13, c2 = $\quad 5$

√

1

○

$c_1 = \frac{-13}{\sqrt{5}}, c_2 = \frac{1}{\sqrt{5}}$c1 = $\quad 5$

√

−13, c2 = $\quad 5$

√

1

○

$c_1 = \frac{13}{\sqrt{5}}, c_2 = \frac{-1}{\sqrt{5}}$c1 = $\quad 5$

$\sqrt{}$
13, c2 = 5

$\sqrt{}$
−1

○
$c_1 = \frac{-13}{\sqrt{5}}, c_2 = \frac{-1}{\sqrt{5}}$c1 = 5

$\sqrt{}$
−13, c2 = 5

$\sqrt{}$
−1

No, the answer is incorrect.
Score: 0
Accepted Answers:
$c_1 = \frac{13}{\sqrt{5}}, c_2 = \frac{-1}{\sqrt{5}}$c1 = 5

$\sqrt{}$
13, c2 = 5

$\sqrt{}$
−1

1 point
Consider an orthonormal basis $\{(1,0,0),(0,1,0)\}$$\{(1,0,0),(0,1,0)\}$ for a subspace WW in R^3R3. If $x = (1,2,3)$x = (1, 2, 3) is a vector in R^3R3, then which of the following represents a vector in WW whose distance from xx is the least? Consider dot product as the standard inner product.

○
(2, 4, 0)(2, 4, 0)

○
(3, 4, 0)(3, 4, 0)

○
(4, 5, 0)(4, 5, 0)

○
(1, 2, 0)(1, 2, 0)

No, the answer is incorrect.
Score: 0
Accepted Answers:
(1, 2, 0)(1, 2, 0)

Level 2

Consider the inner product $\langle a, b\rangle = a_1 b_1 - a_1 b_2 - a_2 b_1 + 4a_2 b_2$ ⟨a, b⟩ = a1b1 − a1b2 − a2b1 + 4a2 b2 where $a = (a_1, a_2), b = (b_1, b_2)$a = (a1, a2), b = (b1, b2) are vectors in R^2R2. Use this information to answer questions 4 and 5.

1 point
Let $x = (1,2)$x = (1, 2). Find the projection of xx in the direction of (3, 4)(3, 4) using the inner product defined above.

○
$\frac{25}{49}$(3, 4)4925(3, 4)

○
$\frac{11}{25}$(3, 4)2511(3, 4)

○

$\frac{25}{49}(1,2)$4925(1, 2)

○

$\frac{11}{25}(1,2)$2511(1, 2)

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\frac{25}{49}(3,4)$4925(3, 4)

1 point

Let x = (1, 2)x = (1, 2), find the projection of xx in a direction perpendicular to (3, 4)(3, 4).

○

$\frac{25}{49}(-4,3)$4925(−4, 3)

○

$\frac{11}{25}(4,\ -3)$2511(4, −3)

○

$(-\frac{26}{49},\ -\frac{2}{49})$(−4926, −492)

○

$(\frac{26}{49},\ -\frac{2}{49})$(4926, −492)

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(-\frac{26}{49},\ -\frac{2}{49})$(−4926, −492)

1 point

Consider an orthogonal basis {(1, 2, 1), (− 2, 0, 2)}{(1, 2, 1), (−2, 0, 2)} of a subspace WW, of the inner product space R^3R3 with respect to the dot product. If y = (1, 2, 3) ∈ R^3y = (1, 2, 3) ∈ R3, then find $Proj_W\,(y)$P rojW (y).

○

$(\frac{1}{3}, \frac{8}{3}, \frac{7}{3})$(31, 38, 37)

○

$(\frac{18}{\sqrt{6}} - \frac{32}{\sqrt{8}}, \frac{36}{\sqrt{8}}, \frac{18}{\sqrt{6}} + \frac{32}{\sqrt{8}})$(6 √ 18 − 8 √ 32, 8 √ 36, 6 √ 18 + 8 √ 32)

○

$(\frac{1}{3},\ -\frac{8}{3}, \frac{7}{3})$(31, −38, 37)

○

$(\frac{18}{\sqrt{6}} - \frac{32}{\sqrt{8}}, \frac{36}{\sqrt{8}},\ -\frac{18}{\sqrt{6}} + \frac{32}{\sqrt{8}})$(6 √

$18 - \sqrt{8}$

$32, \sqrt{8}$

$36, -\sqrt{6}$

$18 + \sqrt{8}$

$32)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(\frac{1}{3}, \frac{8}{3}, \frac{7}{3})(31, 38, 37)$

Check Answers

Your score is: 0/6